HUMEBOT · PROJECT REPORT 6

HumeBot: Classroom Readiness and Institutional Decisions

What has been accomplished toward a university instructional resource, what remains before students could use it, and which decisions belong to whom

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Audience	Faculty, university administrators, educational technologists and researchers in the responsible educational use of AI
Version	1.0
Date	8 November 2026
Project status	Working development application; not deployed, not approved for student use, no qualified model.

The HumeBot project website: https://mdcollier.github.io/humebot-site/

ABOUT THIS REPORT

Central question	What has been accomplished toward making HumeBot available as a university instructional resource, what remains necessary before classroom use, and which decisions belong to the faculty lead and which to the University?
Audience	Faculty, university administrators, educational technologists and researchers interested in the responsible educational use of AI. No knowledge of the project's repository or of the University's internal processes is assumed.
Evidence and scope	The Reading Desk application and its handoff and readiness documentation (October–November 2026), the project's deployment gate and constitutional commitments, the evaluation record summarized in Report 5, and the project's planning records. Technical detail is in Report 1, the educational design in Report 2, the corpus in Report 3, the model research in Report 4.
Status	HumeBot is a working development application tested with fictional accounts and recorded answers. It is not deployed, is not approved for student use, and has no language model qualified for teaching. No formal request for University evaluation or approval has been made, and nothing in this report describes a decision the University has taken.

1. From research prototype to classroom resource

Three things are easily confused when a university project builds software that students might one day use. A **research system** exists to answer questions: it can be run, measured, changed and run again, and its results are its purpose. A **working application** exists to be used: it has an interface, accounts, records and operations, and it can be tested end to end with fictional users and recorded data. An **instructional service approved for student use** is a working application that has, in addition, passed through the decisions that stand between software and students: that the University will host and operate it, that students' data will be handled under rules the University has determined, that what the system shows students meets a standard someone is accountable for, and that an instructor has chosen to use it for a purpose the course has.

HumeBot has been all of the first, has become the second, and is not the third. Its research systems are the subject of Reports 4 and 5. Its working application is the **Reading Desk**: a web application in which a student asks a question about Hume's *Enquiry concerning Human Understanding* and receives HumeBot's explanation beside the passages of Hume it rests on, each verified against a controlled transcription of the 1777 edition, with the whole paragraph, its neighbours and the complete section available to read, and with an instructor able to inspect the evidence, checks and history behind every answer. In a philosophy course its purpose would be modest and specific: to let students ask their own questions of a difficult primary text and be sent back to the text with a reading to test, rather than to an answer to accept.

The project's own standard for the present phase is stated in its records and is the right frame for this report: the aim has been to make deployment and student use *achievable without substantial architectural rework*, with no claim that the application is certified, approved or ready for students. The distance from a working application to an approved instructional service is not a software distance. It is made of evidence the project has not yet produced, operations the University has not been formally asked to provide, and decisions that have not been taken.

ACCOMPLISHED, OUTSTANDING, UNDECIDED

Accomplished. A complete, containerized application with verified text beside evidence-linked interpretation, instructor inspection, role-separated access, append-only records and fail-closed verification, tested end to end in development with synthetic identities, recorded answers and a stand-in model service; a documented handoff for a university cloud team; a register of the requirements that remain.

Outstanding. Everything that needs University services or determinations: hosting, a managed database, sign-in, an approved model service, the written determination on student data, a retention and deletion procedure, accessibility review and operational ownership; and, on the project's side, a model configuration accepted as adequate for students, which none is.

Undecided. Whether a first course will use HumeBot at all, in what configuration, hosted where, with which model provider under which terms; and when to make a formal request to the University.

2. What has been built

The Reading Desk's capabilities are described technically in Report 1; what matters here is what they would do in a course.

Verified passages. Every passage a student sees is built by the software from the controlled text of the 1777 edition and checked, at the moment of display, against that text. The student reads Hume's words, not the model's recollection of them, and can open the surrounding paragraphs and the whole section. A passage that fails verification is withheld, not shown.

Evidence-linked interpretation. HumeBot's explanation is labelled as interpretation and presented sentence by sentence; selecting a sentence shows exactly which words of Hume it rests on. Each sentence carries two separate verdicts: that its source and location are verified, and that its interpretation is not verified as correct. The second verdict is the design's honesty about what software can and cannot establish, and the invitation to the student to judge.

Instructor inspection. For every answer given in their course, an instructor can see what evidence was supplied to the model, what it selected, what was displayed and cited, what the checks found, what the system identified as gaps, and the answer's full history, together with the model and prompt versions that produced it. Instructors can also leave notes that are kept with the answer and cannot be edited or deleted through the application. This is the application's means of educator oversight, and it is built.

Mechanisms for accountable use. Access follows course roles and defaults to deny: students see only their own readings and questions, instructors only their own course's. Every delivery, view, inspection and annotation is recorded, and the records cannot be altered through the application. Students' questions, the model's output and all credentials are kept out of logs. The ability of students to ask new questions is off in every course until its instructor turns it on, and student use as such is refused by the application's configuration until the University's written determination on student data exists and is recorded in it. A new question is answered by a separate worker process that calls the model, re-verifies the result and commits it with its audit records before anything is shown; a failure at any point leaves a recorded state and a notice, never a partial answer.

What is implemented and tested, and what is not. All of the above has been exercised in development: the application runs in containers on a developer's machine, with fictional student and instructor accounts, twenty recorded answers for demonstration, and a local stand-in for the model service that serves recorded outputs and fails closed on anything else. The classroom workflows that the design anticipates have **not** been deployed or evaluated with anyone: a class of students asking live questions through a University sign-in; an instructor reviewing a term's readings; a course sharing and discussing readings in the optional reading commons that exists only as a development build; and the display of an instructor's own lecture material beside HumeBot's readings, which the project has decided upon in principle but not built. The application has never been run on University infrastructure, connected to a University sign-in, backed by a managed database, or attached to any real model service.

3. What remains before students can use it

The project's readiness register lists what stands between the present application and student use. Most items are not technical work for the project; they are services the University would provide or determinations it would make, and the application has been built to fit the common forms of each. In ordinary language, they are these.

Hosting and operations. The application needs a container platform to run its web, worker and migration processes, with a secret store, network ingress and certificates, and someone responsible for monitoring, alerts, log retention, incident response and a freeze on changes during term. The application exposes the health and readiness checks such a platform expects and documents what to watch; none of it has been exercised on University systems.

A managed database. The application stores its records in PostgreSQL and expects a University-managed instance with encrypted connections, separate roles for migration, application and administration, regular backups and, before any student use, a tested restore. The application refuses an unencrypted database connection by configuration.

Sign-in and roles. The application does not manage passwords. It trusts a single identity header set by the University's authenticating gateway, the pattern university single sign-on systems provide, and it needs the University to confirm in writing that the gateway strips any such header a browser might supply, because the application cannot check this itself. Course roles are provisioned by an audited command until the University's source of roles (a roster feed or a group attribute) is known.

A model service. Answering new questions requires a language model reached through an approved service with the settings the pipeline fixes. Only the worker process would call it, over an encrypted connection, sending the student's question and the corpus evidence and never the student's identity. The University would need to approve which service and model may receive student questions and under what terms for retention, logging and training; the project would need to run an authorized compatibility check against that service before any live use. If the approved service uses a different interface or authentication from the one the connector supports, a small adapter would be needed; none is built.

Student data. Before any real student uses the application, a written University determination is needed on four questions the project has deliberately not answered for itself: how student questions and readings are classified; how stored answers and instructor views are treated under student-records law; how long records are kept and how they are deleted, which matters because the application's records are append-only by design and a governed deletion procedure for them does not yet exist; and whether the approved

model service may receive student questions at all. Students would also need to be told what is stored and who can see it. The proposed initial use is an ordinary instructional application, not a research study; any later research use of student records would require its own institutional and ethics determinations.

Accessibility. Automated checks against the WCAG 2.1 AA standard and scripted keyboard tests pass and run as a regression gate in development. That is not a conformance claim. Testing by experienced screen-reader users, real browser zoom, other browsers and assistive technologies, testing with disabled users, an assessment by the University's accessibility office of the hosted application, and the University's own sign-in pages and session behaviour all remain.

Reliability under real conditions. The application's behaviour under real load, real network failures and a real model service has not been observed; its limits on concurrent questions, question length and questions per user per hour are configurable defaults from development, not tuned values.

Four pieces of design work are also identified before live use, none of them an obstacle: live generation must run outside a web request (done, through the worker); a provider adapter may be needed for services that do not use the connector's authentication; retention needs a governed way to delete append-only records; and roles need an agreed provisioning path. The project's assessment is that no architectural barrier to meeting the University's requirements is known. **Technical completion is not authorization.** Every item above that the University would provide or determine is open; none is a decision already taken.

4. Model qualification and scholarly reliability

A working application is not a qualified teaching instrument. The Reading Desk verifies, without exception, that what it shows a student is Hume's text at the place cited. It cannot verify that HumeBot's reading of that text is right, that it preserves Hume's qualifications, or that it omits nothing a careful reader would want. Those are properties of the language model's work, and whether a given model, with the project's prompts and settings, produces readings adequate for students is a question the project treats as requiring its own answer, separate from whether the software works.

The project built the instrument to answer it: a formal **qualification** in which a frozen configuration is run on a large set of newly written questions, sealed until use, and judged against thresholds fixed in advance for delivery, false abstention, serious scholarly error and material omission, with a blinded scholarly review. The one attempt, in October 2026 with an open-weight model on the University's research computing, was executed in full and stopped at its evidence gate before any scholarly review took place: a procedural rule that had never been formally adopted had been invoked, and the frozen procedure did not permit the review to go forward. The attempt established neither a pass nor a fail, and its questions, having been seen, cannot be reused. A redesigned attempt exists only as a plan and would need a fresh set of questions of about the same size, whose supply is a recognized difficulty. Report 5 gives the experimental record. The consequence for the classroom is plain: **no configuration of HumeBot is qualified or accepted for student use**, with any model, open-weight or commercial.

The project's rules require a qualification decision for any live configuration before students see its answers, and treat any change of model, service or settings as a new decision. The formal qualification just described is the instrument the project designed for that decision, and it has not been completed for any configuration. Short of a completed formal qualification, the decision would take the form of the faculty lead's bounded **acceptance review** of live answers to course questions under the project's standards, before any student saw one (Report 1); it is a project governance decision, not a University one, and it

would be reopened whenever the model, service or settings changed. The evidence so far bears directly on what that review would look for. In the project's comparisons the harness's textual guarantees held with every model tried, but interpretive quality varied greatly with the model, and the open-weight configuration's serious errors concerned the scope and finality of Hume's claims. A frontier model produced markedly better readings in a small sample, through a configuration that differs from the frozen one and would itself need acceptance. The project's records also name two fallbacks for a first course, should no live configuration be accepted: a text-only use of the Reading Desk, and a fixed set of reviewed, recorded answers. Both are recorded as fallbacks, not as equivalents of the live, question-centred service the project prefers.

5. Faculty authority and pedagogical use

HumeBot is designed to support a student's engagement with Hume, not to replace the instructor's. The design decisions that carry this are few and deliberate. The instructor decides whether students in a course may ask new questions at all, and may turn the capability off. The instructor sees everything behind every answer and may annotate it. The instructor determines the course's pedagogical objectives, while the application has its own evidence-centred educational design: it answers substantively, shows Hume's text beside the reading and sends the student back to it, and it is committed not to present contested interpretations as settled, a commitment whose full operation awaits curated scholarly data that do not yet exist (Report 3). Instructor materials, lectures, notes, argument reconstructions and course objectives, remain where they are in any course, framing the students' work; they are **not** supplied to the model, and the project has decided that for now they should be displayed as distinct, attributed course artifacts beside Hume's text and HumeBot's readings rather than fed into generation, a presentation that is itself not yet built. Small experiments in November 2026 that did supply an instructor's guidance to the model, under a controlled exception, changed the framing and form of its explanations without reliably improving them (Report 5).

In a course, then, HumeBot's intended role is a reading companion under the instructor's authority: a way for students to ask the text their own questions and to receive a reading they can check against Hume's words, with the instructor's framework governing what the course teaches and the instructor able to see how the tool is being used. The instructor's feedback on answers is intended, in the longer term, to become the principal source of improvement to the model's behaviour, through an interface the project has specified in outline and not built, and only under University governance of the data involved.

What this would do for students' learning is not known. **No student has used HumeBot, and no learning outcome has been studied.** The project's evaluations support claims about the reliability of the text students would see; they support no claim about educational benefit, and the project has resolved not to infer one from the design (Report 2). A study of whether the Reading Desk helps students read Hume would be research, requiring the University's research governance and students' consent, and would follow, not precede, an ordinary instructional deployment that had itself been approved.

6. Decisions requiring institutional involvement

The decisions that remain divide between the project and the University, and the table below separates what the project has prepared, what the University would need to evaluate, and what is unresolved. Nothing in it records a decision the University has taken. No formal request for University evaluation or

approval has been made, and the project's own view is that none should be made until its remaining decisions are taken and a specific course use is proposed.

Table 1. Decisions before classroom use: what the project has prepared, what the University would evaluate, and what remains unresolved.

Area	Prepared by the project	For the University to evaluate	Unresolved
Hosting and operations	A single container image with health and readiness checks, documented configuration, smoke tests needing no student data, and a handoff for a cloud team	Which platform; secrets, ingress and certificates; monitoring and alerting; incident ownership; change freeze in term	Who would operate the service, and on what terms
Database	Standard PostgreSQL use: separate roles, append-only records, encrypted connections required	A managed instance; backups; a tested restore; encryption at rest; migration procedure	Retention schedule; a governed deletion procedure for append-only records (not built)
Sign-in and roles	Trusted-header identity, default-deny roles, audited provisioning	Which gateway; the stable identifier; written confirmation that client-supplied identity headers are stripped; the source of course roles; session rules	Whether roles come from a roster feed or group attributes
Model service	A connector with the pipeline's fixed settings and a served-model check; an authorized compatibility probe	Which service and model may receive student questions; retention, logging and training terms; authentication; quotas and cost	External API or University-hosted open-weight model: the project's working assumption is the former; the choice is open, and no model is selected
Student data and privacy	Minimal records; no raw model output or prompt text stored; identity never sent to the model; student use refused until a determination is recorded	Classification of student questions and readings; treatment under student-records law; retention; notice to students; provider approval	The written determination itself
Accessibility	Automated WCAG 2.1 AA checks and keyboard tests as a regression gate; a register of open items	Assessment of the hosted application; sign-in pages and session behaviour; testing with assistive-technology users	Several judgement items the project has flagged for the accessibility office
Scholarly adequacy of answers	A qualification design with fixed thresholds; an acceptance procedure for live configurations	— (a project responsibility, informed by the model service the University approves)	No configuration qualified or accepted; a fresh qualification set would be needed
Instructional use	Instructor controls over live questions and reading view; the Evidence Inspector; the display-only decision on instructor materials	Whether a course may use the service, under what notice to students and what oversight	Which course, when, in which configuration
Research use	None proposed for the initial use; the application's research schema is created empty	Any later research would need research governance and consent	Not proposed

The project's unresolved decisions are its own to make and are stated as such: whether to pursue an external inference service or a University-hosted open-weight model for a first course, a question on which its records show a working assumption but no ruling; what configuration to put forward for acceptance, and when; how to design a deletion procedure consistent with append-only records; and which course would be first. The University's decisions would begin when the project makes a specific proposal.

7. What classroom readiness would mean

A responsible transition from research to student use would not be a software release. It would be the coincidence of four conditions, each with a different owner.

A **functioning application**, operated by the University on its infrastructure, connected to its sign-in and managed database, with operations agreed and a restore tested: the project has built the application and documented the rest; the University has not been formally asked to provide it.

An **appropriately qualified model**: a specific configuration of model, service and settings for which the project has made the qualification decision described in section 4, re-reviewed whenever it changes. None exists; the formal attempt did not reach review, and no acceptance review has been held.

Institutional authorization: the University's written determination on student data, its approval of a model service that may receive students' questions, its accessibility review, and its agreement that a particular course may use the service. No formal request has been made.

Educational oversight: an instructor who has chosen to use the tool for a purpose the course has, who controls whether and how students use it, who can inspect what it does, and whose framework, not the model's, governs what the course teaches. The application is built for this; the first such instructor would be the faculty lead.

The project's planning names a first course in 2027 as a target, explicitly subject to qualification and institutional approval, and states that no live configuration is declared achievable. Building a research application that keeps a language model accountable to a primary text, and testing it end to end, is a real achievement, documented in the five preceding reports. Making it a classroom resource requires evidence about the adequacy of any model's readings that the project has not yet produced, operational preparation only the University can provide, and institutional decisions not yet taken. Readiness is the name for all four together: the project has built the application component, and the other three remain to be met.

Sources and document status

- Reading Desk deployment-readiness register, pathway assessment (29 October 2026), technical handoff and model-service contract; Reading Desk README, release-candidate acceptance record (30 October 2026) and reading commons development report (3 November 2026).
- The deployment gate on student data (Workbench specification §9); the constitutional document §3; the closure of the instructor-guidance research arc (5 November 2026); strategic directions §14.6, §15.2 and §16; the Instructional Qualification 0.1 decision record (closure); Report 5.

Document status. Version 1.0, 8 November 2026. A faculty-led project report of the University of Minnesota Morris; not an official University publication and not University policy. HumeBot is a development prototype: not deployed, not approved for student use, and not qualified; no formal request for University evaluation or approval has been made.